Quantifying Effect Modifiers Similarity for Regional Pooling Strategy in Multi-Regional Clinical Trials

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Outline

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2. **Methods**: nABCD definition, theoretical foundation, and clinical calibration
3. **Simulation**: Estimation performance across 7 scenarios
4. **Application**: Hypothetical thrombolytic MRCT using GUSTO-I
5. **Discussion**: Key findings and future work

Background and Objectives

Background

- Treatment effect consistency across regions is assessed in MRCTs
- Individual regional sample sizes are often too small alone for consistency assessment
- **ICH E17 strategy**: pool regions with similar **effect modifier (EM)** distributions
- At the **planning stage**, sponsors must select pooling partners with evidence that regions are "similar enough"

Objectives

- Develop a **quantitative index** that captures distributional similarity
- Deliver a **practitioner-facing tool** for the planning stage of an MRCT

Methods

Dissimilarity Index: nABCD

Wasserstein-1 distance

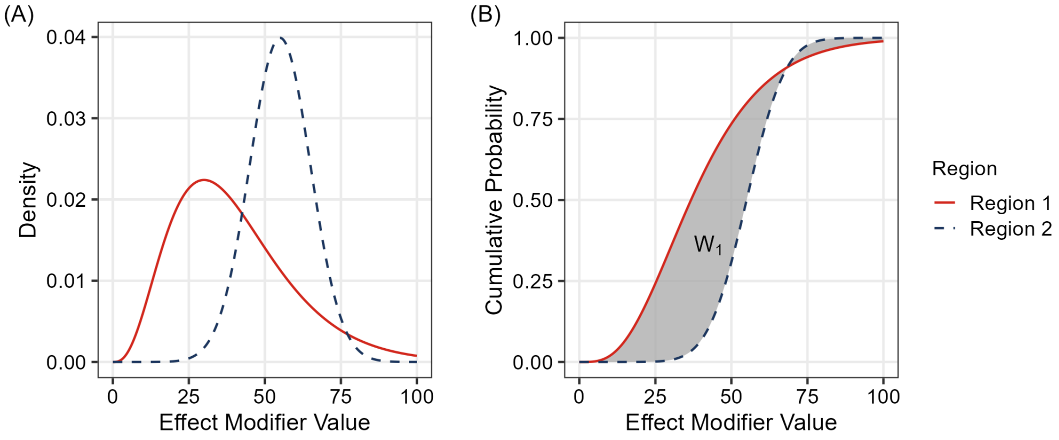
$$W_1(F_1, F_2) = \int |F_1(x) - F_2(x)|\, dx$$

The total area between two CDFs — captures **location, scale, and shape** differences.

nABCD: normalized Area Between Cumulative Distributions

$$\text{nABCD}(F_1, F_2) = \frac{W_1(F_1, F_2)}{\text{IQR}_{\text{pooled}}}$$

Scale-free dissimilarity; a 1-IQR location shift yields nABCD = 1.0.



Heterogeneity Bound

Kantorovich–Rubinstein Duality

$$W_1(F_1, F_2) = \sup_{\|f\|_{\mathrm{Lip}} \leq 1} \left| \int f\, dF_1 - \int f\, dF_2 \right|$$

Heterogeneity Bound

If the CATE $\tau(x)$ is Lipschitz continuous with constant L:

$$|\bar{\tau}_1 - \bar{\tau}_2| \leq L \cdot \mathrm{IQR}_{\text{pooled}} \cdot \mathrm{nABCD}(F_1, F_2)$$

- L: upper bound on treatment effect change per unit change in the EM
- **Theoretical bridge** from distributional distance to treatment effect differences

Estimation

Empirical estimator

$$\widehat{\text{nABCD}} = \frac{\sum_{k=1}^{n_1+n_2-1} |\hat{F}_1(x_{(k)}) - \hat{F}_2(x_{(k)})| \cdot (x_{(k+1)} - x_{(k)})}{\widehat{\text{IQR}}_{\text{pooled}}}$$

Inference: Percentile bootstrap

- Asymptotic distribution non-standard: $\sqrt{n}\, W_1(\hat{F}_n, F) \xrightarrow{d} \int |B(F)|\, dx$ — Brownian bridge (del Barrio 1999)
- No universal critical values → **percentile bootstrap** ($B = 2{,}000$ resamples)
- Boundary case $F_1 = F_2$: parameter space edge — addressed in simulation

Clinical Calibration

Pathway 1 — When L is available: $\Delta_{\max}$

$$\Delta_{\max} = L \cdot \mathrm{IQR}_{\mathrm{pooled}} \cdot \mathrm{nABCD}$$

Worst-case regional treatment effect difference on the **clinical scale**.

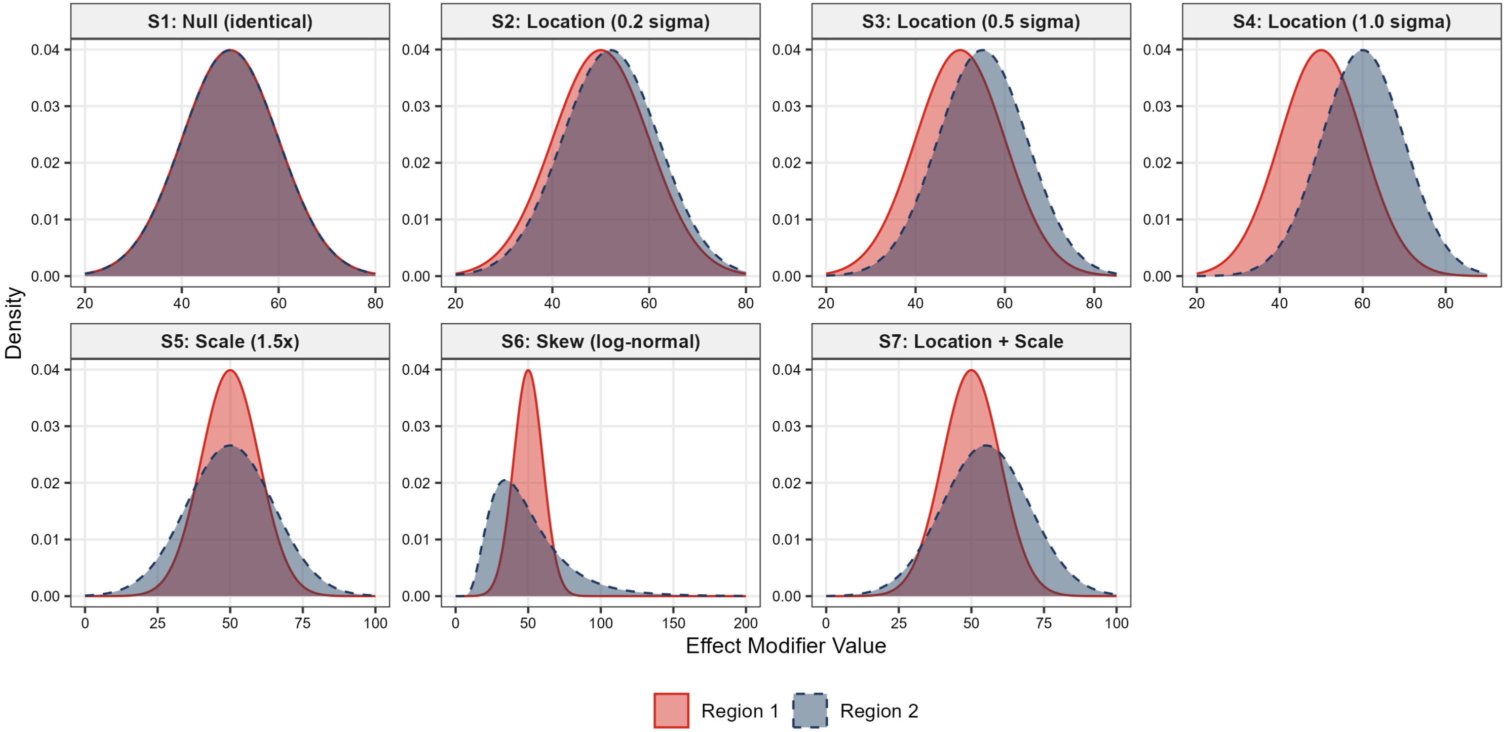
Pathway 2 — When L is unknown: L^*

$$L^* = \frac{\Delta_{\mathrm{clin}}}{\mathrm{IQR}_{\mathrm{pooled}} \cdot \mathrm{nABCD}}$$

- Δ_{clin}: clinically important difference
- L^* above plausible range $\Rightarrow$ distributional difference unlikely to be clinically concerning

Simulation Design

- Distribution pairs for scenarios (**S1–S7**) covering:
  - Null (S1), location-shift (S2–S4), scale-shift (S5), skewness (S6), combined location+scale (S7)
- **10,000 replications** per scenario × sample size cell
- Sample sizes: **n = 50, 100, 200** per region
- Bootstrap **B = 2,000**, percentile 95% CIs



Simulation Results

Bias

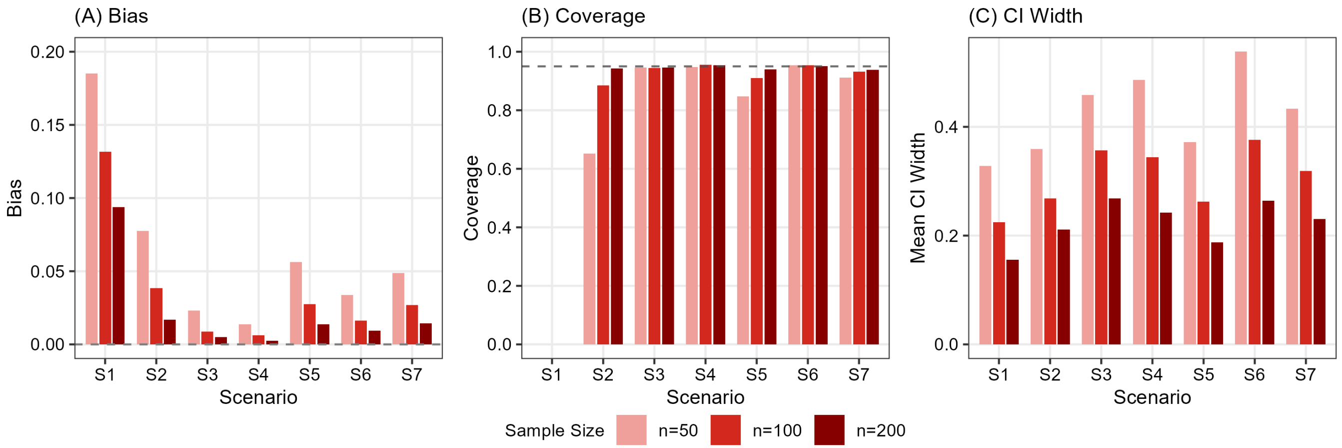
Bias is **negligible** for non-null scenarios (S3–S7) at $n \geq 100$. At the null (S1), a positive bias persists because the Wasserstein distance is **non-negative**.

Coverage

95% bootstrap CI coverage is **0.88–0.96** for non-null scenarios (S3–S7) at $n \geq 100$. At the null (S1), the non-negative bootstrap CI cannot contain zero.

CI Width

CI width **shrinks with sample size** as expected.



Application: GUSTO-I

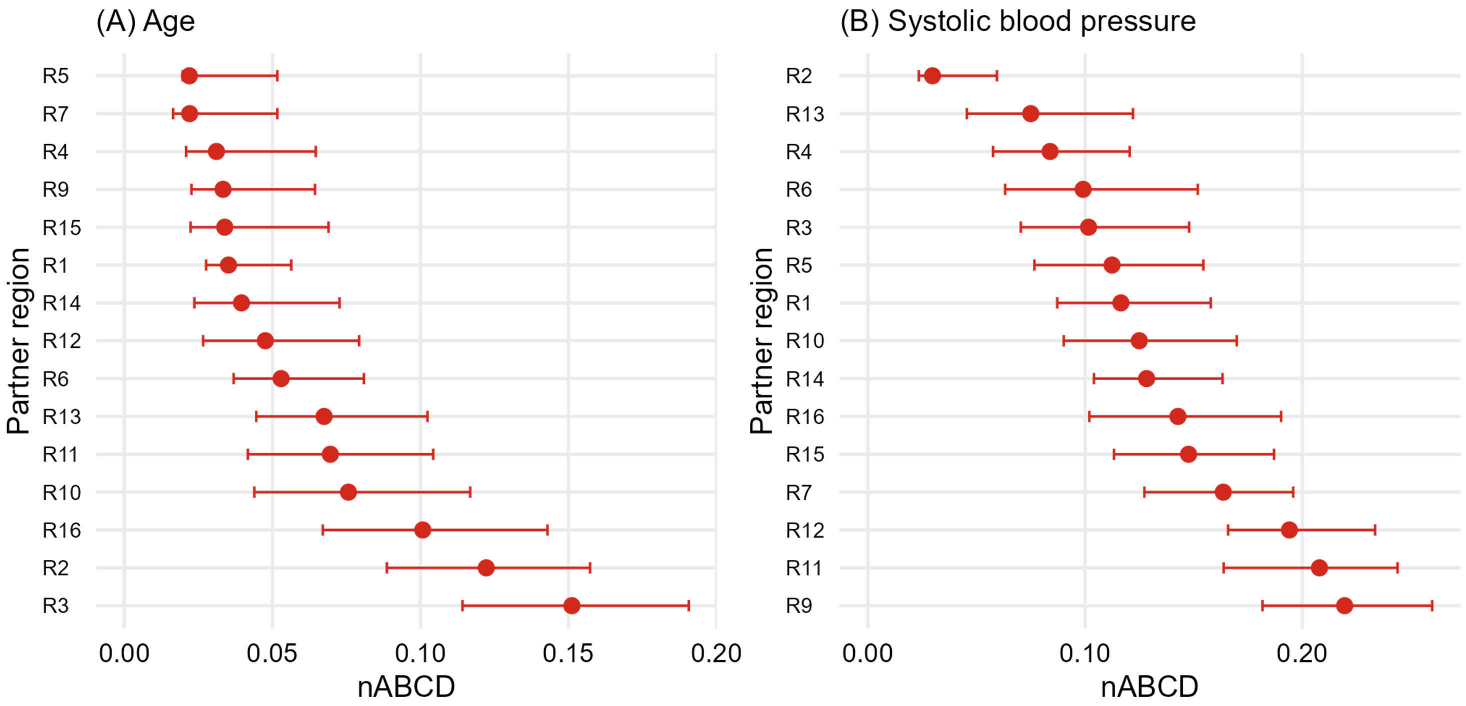
Goal

A sponsor planning a Phase 3 MRCT for a novel **thrombolytic** in acute myocardial infarction (AMI) must identify pooling partners for a region within a regulatory market. **Age** and **systolic blood pressure (SBP)** are candidate EMs. The goal is to identify regions with similar EM distributions to the anchor region (**Region 8**).

Data

- **GUSTO-I** individual patient data (**16 anonymized regions**) serves as the distributional source
- Anchor: Region 8; remaining 15 regions evaluated as candidate pooling partners

Application: Joint Eligibility via L*



- **Joint eligibility** at $\Delta_{\text{clin}} = 1\%\text{pt}$ with class-level upper bounds ($L_{\text{UB,age}} = 10^{-2}$/yr; $L_{\text{UB,SBP}} = 2\times10^{-3}$ /mmHg): **6 partners (R1, R4, R5, R6, R14, R15)** emerge as jointly eligible on both EMs
- **R4** is the leading candidate — balanced ranking on both EMs

Findings

1. **Methodological gap filled**: nABCD provides the **quantitative metric** for EM distributional similarity — transforming "sufficiently similar" from qualitative judgement into a reproducible quantitative basis.
2. **Heterogeneity bound**: Via Kantorovich–Rubinstein duality, the maximum CATE difference is bounded — providing the theoretical bridge from distributional distance to treatment effect differences.
3. **Reliable bootstrap inference at moderate n**: Across 7 simulation scenarios, small bias and adequate coverage for non-null distributional differences at $n \geq 100$.
4. **Dual-pathway clinical calibration**: When L is estimable, $\Delta_{\max}$ gives clinical-scale interpretation; when L is unknown, L^* reverse-calculation supports judgement without imposing a fixed nABCD cutoff.

Future Work

1. **Categorical EM extension**
   Current framework focuses on continuous EMs; extension to binary and multi-level categorical EMs requires distinct distance metrics designed for discrete distributions.
2. **Multivariate extension**
   EM evaluation could be extended to joint multivariate distributional comparison via Wasserstein-2 or sliced Wasserstein distances, capturing correlation structure.
3. **Upstream EM identification**
   Which baseline characteristics constitute relevant EMs is a non-trivial pre-step assumed solved in this work; formal methodology deserves dedicated investigation.